Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/16/2008
PI Name:	Menaker, Michael Ph.D.		
Project Title:	A Model of Circadian Disruption in the Space Environment		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Charlottesville	**State:**	VA
Zip Code:	22904-4328	**Congressional District:**	5
Comments:	Deceased as of February 2021		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	07/01/2004	**End Date:**	06/30/2008
No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	9
No. of Bachelor's Candidates:	14	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Yoshikawa, Tomoko (University of Virginia)		
Grant/Contract No.:	NCC 9-58-HPF00406		
Performance Goal No.:			
Performance Goal Text:			

(1) ORIGINAL AIMS OF PROJECT
SPECIFIC AIM I: Using rhythms of gene expression reported by a luciferase transgene, we will identify the signals linking brain and peripheral oscillators.

SPECIFIC AIM II: We will evaluate potential countermeasures by exposing our experimental animals to conditions that disrupt the normal circadian phase map and then determining if the potential countermeasures are able to restore it rapidly.

SPECIFIC AIM III: Investigate the effects of chronic low dose methamphetamine in synchronizing disrupted rhythms and evaluate related drugs as potential countermeasures.

Task Description:	(2) THE KEY FINDINGS OF THE PROJECT We have extended our studies of the methamphetamine-sensitive circadian oscillator (MASCO) and have discovered a sex difference in the effect of methamphetamine on C3H mice, i.e., females are much less responsive than are males. We have now demonstrated that gonadectomy does not alter the circadian response of either males or females to methamphetamine, and thus sex hormones are unlikely to explain the sex difference. In addition, we have been screening circadian mutant mice to determine if there is a role for canonical clock genes in the effect of methamphetamine on free-running period length. All of the arrhythmic mutant mice screened to date have become rhythmic in the presence of methamphetamine. This demonstrates that MASCO is indeed a distinct brain oscillator which may be involved in the pathological responses of the circadian system to the space environment. We have also extended our study of the phase synchronizing effect of methamphetamine on circadian oscillators. We previously demonstrated that ablation of the SCN results in phase desynchrony between internal oscillators and that methamphetamine administration is capable of restoring coherent phase relationships between peripheral tissues. In the past year we have run another group of animals confirming and elaborating on our prior experiments. We have now included two additional tissues (lung and pituitary) in our analysis and compared the effect of methamphetamine to that of food restriction. The results demonstrate that both methamphetamine and food restriction can restore phase synchrony, in at least some tissues, following SCN lesion. This is particularly exciting because programming meal times is a non-invasive and potentially beneficial way to treat circadian disorganization produced by the space environment. We have finished our studies of the phase-controlling signals to two internal organs, the ovary and the submaxillary gland. One paper has been published from these data and another is in preparation. To summarize these findings: the ovary responds to pituitary hormones and does not require neural signals; the salivary gland responds to both neural signals and to feeding time, if and only if the neural inputs are cut. This variety of responses suggest that internal circadian organization is normally maintained by many different signals that are specific to particular organs or groups of organs and may be hierarchically organized. Understanding and eventually controlling them will be an important but difficult undertaking. (3) THE IMPACT OF THESE FINDINGS ON THE HYPOTHESES, OBJECTIVES AND SPECIFIC AIMS OF THE ORIGINAL PROPOSAL Our project is intended to provide some of the basic data that will be needed to minimize the effects of disrupting the circadian rhythms of astronauts and ground-based support personnel during their missions. Such disruption is an inevitable result of the work schedules that these men and women are required to follow. Disruption of circadian organization, which includes but is not limited to sleep deprivation, has repeatedly been shown to affect alertness, cognition, and performance on a variety of tasks. In order to minimize these effects, we have to understand the organization of the system that is being affected. In particular, we have to identify general signals such as meal timing that are capable of maintaining internal synchrony with minimal associated side effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There are several potential earth-based applications of our work that could generally improve human health: • Methylphenidate and, more importantly, rigid feeding schedules may be useful for temporarily stabilizing circadian phase in the face of disrupting environments. • Sympathetic nervous system agonists and antagonists of several kinds may be used to modify phase relationships among some peripheral circadian oscillators and thus improve the outcomes of chronic therapeutic treatments of cancer and other chronic conditions (i.e., high blood pressure and irregular heart beat). • The reproductive cycles--in particular their circadian components--of women engaged in activities with disruptive schedules may be regularized by carefully timed application of LH or FSH. • Our data suggest that shift work schedules in which cycles are repeatedly delayed will be less damaging to the health of workers than those in which these cycles are repeatedly advanced.
Task Progress:	• We have shown that the circadian phase of the ovary does not respond to neural signals but does respond to pituitary hormones (LH and FSH). We have obtained preliminary data supporting the hypothesis that the circadian oscillator in the ovary influences the timing of ovulation. This is a reasonable hypothesis given our findings, but runs counter to the strongly held prevailing idea that the timing of ovulation is exclusively controlled by the pituitary. • We have shown that disrupted internal circadian organization can be restored by chronic application of low dose methamphetamine and, more importantly, by rigid timing of meals. • We have demonstrated that the methamphetamine-sensitive circadian oscillator does not rely on the canonical molecular circadian machinery.
Bibliography Type:	Description: (Last Updated: 03/07/2011)
Articles in Peer-reviewed Journals	Yoshikawa T, Sellix M, Pezuk P, Menaker M. "Timing of the ovarian circadian clock is regulated by gonadotrophins." Endocrinology. 2009 Sep;150(9):4338-47. Epub 2009 Jun 11. PMID: 19520783 , Sep-2009
Articles in Peer-reviewed Journals	Kwak Y, Lundkvist GB, Brask J, Davidson A, Menaker M, Kristensson K , Block GD. "Interferon-gamma alters electrical activity and clock gene expression in suprachiasmatic nucleus neurons." J Biol Rhythms. 2008 Apr;23(2):150-9. PMID: 18375864 , Apr-2008
Articles in Peer-reviewed Journals	Vujovic N, Davidson AJ, Menaker M. "Sympathetic input modulates, but does not determine, phase of peripheral circadian oscillators." Am J Physiol Regul Integr Comp Physiol. 2008 Jul;295(1):R355-60. PMID: 18434440 , Jul-2008
Articles in Peer-reviewed Journals	Sellix MT, Yoshikawa T, Menaker M. "A circadian egg timer gates ovulation." Curr Biol. 2010 Mar 23;20(6):R266-7. PMID: 20334830 , Mar-2010

Awards	Menaker M. "Lerner's Frog Award for lifetime achievement in pineal research at the Gordon Conference on Pineal Cell Biology, Barga, Italy, April, 2008." Apr-2008